T_0 T_1 .

(3)

$$(T_1 - T_0) = \gamma (T_0 - T_x), \quad (4)$$

γ .

γ

$$(T_2 - T_1) = \gamma (T_1 - T_x). \quad (5)$$

(4) (5)

$$\frac{(T_0 - T_x)}{(T_1 - T_0)} = \frac{(T_1 - T_x)}{(T_2 - T_0)}, \quad (6)$$

$$T_x = \frac{T_0 T_2 - T_1^2}{(T_0 + T_2) - 2T_1}. \quad (7)$$

$T_0 = 373\text{K}$, 10

: $T_1 = 337K$, 20 $T_2 = 319K$.

(7)

$$T_x = 301K = 28°C. \quad (8)$$

평가 항목	배점
	1.5
: $T_x = 27.5 - 28.5°\text{C}$	1.5
: $T_x = 27.0 - 29.0°\text{C}$	(1.0)
: $T_x = 26.5 - 29.5°\text{C}$	(0.5)
	0
합계	**3.0**

문제 1C (3.0점)

( ) R, X .

.

$$I = \frac{U}{\sqrt{(r+R)^2 + X^2}}.$$

$$P = \frac{1}{2} I^2 R = \frac{U^2 R}{2[(r+R)^2 + X^2]}.$$

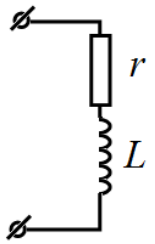
$X = 0$. ,

. $R = r$.

.

$\frac{1}{\omega C} = \omega L$ $L = \frac{1}{\omega^2 C} = 1.00 \cdot 10^{-2}\ Hn.$

2019 , $1.00 \cdot 10^{-2}\ Hn$.



$$P_{max} = \frac{1}{2}\frac{U^2}{4r} = \frac{U^2}{8r} = 13.9\ \text{mW}.$$

평가 항목	배점
	1